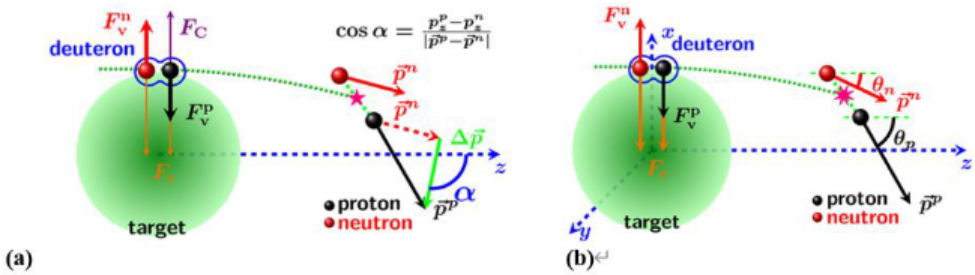
Techniques for measurement of spin-1 polarization P_{zz}

Baiting Tian

(Dated: May 26, 2023)

Abstract

The nuclear matter equation of state has always been a key area in the field of nuclear physics. The determination of the symmetry energy term is particularly important. In the saturation density region, many ground experiments can determine the parameters of the symmetry energy. However, research in the non-saturation density region is still lacking, and the symmetry energy cannot be well determined in the non-saturation region. The dynamic isovector reorientation of deuteron is a phenomenon in which the neutron and proton distribution angles after the deuterium nucleus breaks up due to isovector force caused by the target nucleus scattering are different. We plan to use the isospin polarization effect of the deuterium nucleus to explore the symmetry energy at 0.5 times the saturation density. This article is the preliminary preparation before the experiment, including the monitoring and reconstruction of the polarization state of the deuterium nucleus.



I. THE DYNAMIC ISOVECTOR REORIENTATION OF DEUTERON

The equation of state (EOS) of isospin asymmetric nuclear matter is still a hot topic nowadays. Especially, the symmetry energy which characterizes isospin dependence of EOS has received considerable attention in recent year. Due to the difficulty in solving quantum many-body problems theoretically and the lack of experimental data, there is still significant uncertainty in the symmetry energy in the non-saturated density region. Therefore, we need more accurate probes for the symmetry energy.

The isovector reorientation effect has been reported as a clean and sensitive probe to $E_{sym}(\rho)$ at about $0.5\rho_0$. In the peripheral scattering of a polarized deuteron beam on a heavy target, the isovector force, being attractive to proton and repulsive to neutron, acts like a torque on the deuteron and causes an additional rotation. As a result, the following breakup carries the effect of the rotation, which is sensitive to the isovector force. Detailed simulations demonstrate that the angular distribution of the correlated neutron and proton from the deuteron breakup is very sensitive to the isovector potential, but insensitive to the isoscalar nuclear potential. The experimental observable is quite simple to achieve: one can measure the angular distribution of the relative momentum vector of the neutron and the proton, defined as α, or the angular distribution difference of neutron and proton, defined as $\Delta\theta$.

Essentially, from the view of optical model, the angular distribution difference of the neutron and the proton from the deuteron breakup reflects indeed the difference of the n-A and p-A optical potential. According to Hugenholtz–Van Hove theorem, the analytical relation of the symmetry energy and the globe optical potential can be derived.

II. MONITOR THE TENSOR POLARIZATION p_{zz}

A. Density matrix

In quantum dynamics , we can use a vector in Hilbert space to describe the pure state. And we have to use the density matrix for mixed state which deal with a statistical ensemble of possible preparations.

For a system with probility p_j in $|\phi\rangle$, the measurement is the average of the individual pure state values, weighted by their probabilities:

$$\begin{aligned}\langle\hat{f}\rangle &= \sum_j p_j \left\langle\psi_j|\hat{f}|\psi_j\right\rangle \\ &= \sum_j p_j \operatorname{tr}\left(\left|\psi_j\right\rangle\left\langle\psi_j\right|\hat{f}\right) \\ &= \operatorname{tr}\left(\sum_j p_j \left|\psi_j\right\rangle\left\langle\psi_j\right|\hat{f}\right) \\ &= \operatorname{tr}(\rho\hat{f}).\end{aligned} \tag{1}$$

The density matrix is :

$$\rho = \sum_i p_i \left|\phi\right\rangle \left\langle\phi\right| \tag{2}$$

A pure state $\phi = \sum c_i\phi_i$ is unable to describe an unpolarized state. Take the example of a spin-$\frac{1}{2}$ system, based on symmetry, it should have a random orientation. However, a state like $\phi = c_1\left|+\right\rangle + c_2\left|-\right\rangle$, even though it has an equal probability of exploring the spin-up and spin-down eigenstates upon measurement when $c_1 = c_2 = \frac{1}{\sqrt{2}}$, it has a definite spin direction. The polar and azimuthal angles are determined by $\frac{c_1}{c_2} = \frac{\cos(\beta/2)}{e^{i\alpha}\sin(\beta/2)}$.

B. The Analysing Power

Deuteron is a particle with spin 1. The Cartesian components of the spin operator are:

$$S_x = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad S_y = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -\mathrm{i} & 0 \\ \mathrm{i} & 0 & -\mathrm{i} \\ 0 & \mathrm{i} & 0 \end{pmatrix} \quad S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{3}$$

The density matrix of a spin-1 particle can be decomposed using Pauli matrices and products of Pauli matrices. This set of matrices is called the rectangular operators. The rectangular operators for a spin-1 particle are as follows:

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4}$$

$$\mathscr{P}_x = S_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tag{5}$$

$$\mathscr{P}_y = S_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -\mathrm{i} & 0 \\ \mathrm{i} & 0 & -\mathrm{i} \\ 0 & \mathrm{i} & 0 \end{pmatrix} \tag{6}$$

$$\mathscr{P}_z = S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \tag{7}$$

$$\mathscr{P}_{xy} = 3S_x S_y = \frac{3\mathrm{i}}{2} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \tag{8}$$

$$\mathscr{P}_{xz} = 3S_x S_z = \frac{3\mathrm{i}}{\sqrt{8}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \tag{9}$$

$$\mathscr{P}_{yz} = 3S_y S_z = \frac{1}{\sqrt{8}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} \tag{10}$$

$$\mathscr{P}_{xx} = 3S_x S_x - 2I = \frac{1}{2} \begin{pmatrix} -1 & 0 & 3 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{pmatrix} \tag{11}$$

$$\mathscr{P}_{yy} = 3S_y S_z - 2I = \frac{1}{2} \begin{pmatrix} -1 & 0 & -3 \\ 0 & 2 & 0 \\ -3 & 0 & -1 \end{pmatrix} \tag{12}$$

$$\mathscr{P}_{zz} = 3S_z S_z - 2I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{13}$$

All the matrices in 13 are constructed to be traceless except for the identity matrix. As a result, these traceless matrices are orthogonal to the identity matrix, and their expectation values in an unpolarized state are all zero. These ten operators can span a space that can fully describe the density matrix. However, the last three operators are linearly dependent14. In fact, only nine operators are needed to form a complete basis set.

$$\mathscr{P}_{xx} + \mathscr{P}_{yy} + \mathscr{P}_{zz} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{14}$$

We can expand density matrix as follows:

$$\begin{aligned} \rho = \frac{1}{3}\{I + \frac{3}{2}\left(p_x\mathscr{P}_x + p_y\mathscr{P}_y + p_z\mathscr{P}_z\right) + \frac{2}{3}\left(p_{xy}\mathscr{P}_{xy} + p_{yz}\mathscr{P}_{yz} + p_{xz}\mathscr{P}_{xz}\right) \\ + \frac{1}{6}(p_{xx} - p_{yy})(\mathscr{P}_{xx} - \mathscr{P}_{yy}) + \frac{1}{2}p_{zz}\mathscr{P}_{zz}\} \end{aligned} \tag{15}$$

where $p_i = Tr(\rho\mathscr{P}_i)$

We can also rewrite it in the symmetric form:

$$\begin{aligned} \rho = \frac{1}{3}\{I + \frac{3}{2}\left(p_x\mathscr{P}_x + p_y\mathscr{P}_y + p_z\mathscr{P}_z\right) + \frac{2}{3}\left(p_{xy}\mathscr{P}_{xy} + p_{yz}\mathscr{P}_{yz} + p_{xz}\mathscr{P}_{xz}\right) \\ + \frac{1}{3}(p_{xx}\mathscr{P}_{xx} + p_{yy}\mathscr{P}_{yy} + p_{zz}\mathscr{P}_{zz})\}. \end{aligned} \tag{16}$$

Since quantum mechanics is a linear theory, the final state and initial state can be connected by the scattering matrix M:

$$|\phi_f\rangle = M\,|\phi_i\rangle\,, \tag{17}$$

So , the relationship between final system's density matrix and the initial system's density matrix is :

$$\rho_f = \sum p_i\,|\phi_f\rangle\,\langle\phi_f| = \sum p_i M\,|\phi_i\rangle\,\langle\phi_i|\,M^\dagger \tag{18}$$

$$= M\rho_i M^\dagger \tag{19}$$

For unpolarized deuteron, we can write the density marix:

$$\rho_{0i} = 1/3 \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \tag{20}$$

where $\frac{1}{3}$ is for the normalization requirement.,subscript "0" represents the unpolarized state. The final state is:

$$\rho_{0f} = M\rho_i M^\dagger = \frac{1}{3} M M^\dagger. \tag{21}$$

The differential scattering cross-section for the unpolarized deuteron is given by

$$I_0(\theta) = \frac{1}{3} Tr M M^\dagger \tag{22}$$

And for polarized case , the density matrix is:

$$\begin{aligned} \rho_i = \frac{1}{3}\{I + \frac{3}{2}\left(p_x\mathcal{P}_x + p_y\mathcal{P}_y + p_z\mathcal{P}_z\right) + \frac{2}{3}\left(p_{xy}\mathcal{P}_{xy} + p_{yz}\mathcal{P}_{yz} + p_{xz}\mathcal{P}_{xz}\right) \\ + \frac{1}{3}(p_{xx}\mathcal{P}_{xx} + p_{yy}\mathcal{P}_{yy} + p_{zz}\mathcal{P}_{zz})\} \end{aligned} \tag{23}$$

The final system's density matrix is:

$$\begin{aligned} \rho_f = M\frac{1}{3}\{I + \frac{3}{2}\left(p_x\mathcal{P}_x + p_y\mathcal{P}_y + p_z\mathcal{P}_z\right) + \left(p_{xy}\mathcal{P}_{xy} + p_{yz}\mathcal{P}_{yz} + p_{xz}\mathcal{P}_{xz}\right) \\ + \frac{1}{3}(p_{xx}\mathcal{P}_{xx} + p_{yy}\mathcal{P}_{yy} + p_{zz}\mathcal{P}_{zz})\}M^\dagger \end{aligned} \tag{24}$$

If we define the analysing power as :

$$A_i = M\mathcal{P}_i M^\dagger \tag{25}$$

$$A_{ij} = \frac{Tr(M\mathcal{P}_{ij}M^\dagger)}{Tr(MM^\dagger)} \tag{26}$$

we can write the relationship between unpolarized and polarized differential scattering cross-section :

$$\begin{aligned} I &= \frac{1}{3}\{MM^\dagger + MM^\dagger\frac{3}{2}\sum p_i A_i + MM^\dagger\frac{1}{3}\sum_{i\neq j} p_{ij}A_{ij}\} \\ &= I_0(\theta)\{1 + \frac{3}{2}\sum_i p_i A_i + \frac{1}{3}\sum_{ij} p_{ij}A_{ij}\}. \end{aligned} \tag{27}$$

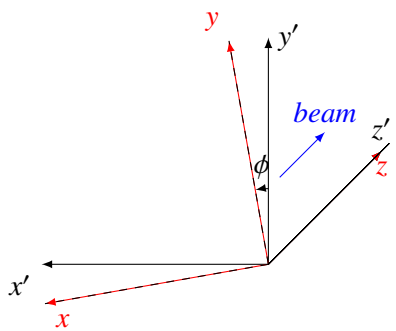


FIG. 1: coordinate system

We will now describe the coordinate system we have chosen, as shown in Figure 1. The S coordinate system is the projectile coordinate system, where its z-axis aligns with the direction of the incident particle k_{in}, and the y-axis is perpendicular to both the incident particle direction and the outgoing particle direction, which is along $k_{\text{in}} \times k_{\text{out}}$. In the beam frame S', the z' axis aligns with the incident particle direction k_{in}, and the y' axis points upwards. In Figure 1, ϕ represents the azimuthal angle. The motion direction of the outgoing particle in S' is given by $(\sin\theta\sin\phi, \sin\theta\cos\phi, \cos\theta)$.

C. Symmetry analysis

In the S system, we analyze the properties of the analyzing power A_i. In this coordinate system, A_i depends only on θ and not on ϕ, which makes our analysis convenient.

1. Restrictions from Parity Conservation

Under parity transformation, except for pseudovectors like spin, all vectors reverse direction. Therefore, $k_{\text{in}} \to -k_{\text{in}}$, $k_{\text{out}} \to -k_{\text{out}}$, and $k_{\text{in}} \times k_{\text{out}} \to k_{\text{in}} \times k_{\text{out}}$. Since x and z are linear combinations of k_{in} and k_{out}, they change sign under parity transformation. However, the y-axis remains unchanged. If parity is conserved, then the transformed system should be identical to the original system. This implies that all coefficients, including those related to the analyzing power, should remain the same before and after the system transformation.

Parity conservation requires that an observable must have an even $N_x + N_z$ for non-zero values.

Here, N_x and N_z represent the number of occurrences of x and z as subscripts, respectively. The subscripts actually represent the components of a tensor. When $N_x + N_z$ is odd, the components of the tensor change sign under coordinate transformation due to parity, but due to the restriction from parity conservation, the observable can only be zero.

D. Coordinate System Transformation

In the projectile coordinate system S, we can conveniently investigate the properties of the analyzing power. However, our focus is on the polarization state in the beam frame S'. Therefore, we need to express the vector p in S coordinate system in terms of the vector p' in S' coordinate system. As shown in Figure 1, S is obtained from S' by rotating around the negative z'-axis by an angle ϕ. It is straightforward to express the vector transformation as follows:

$$\begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = \begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{pmatrix} \tag{28}$$

We can write it as $\overrightarrow{p} = U\overrightarrow{p'}$. The transformation of a second-order tensor can be written as $\overleftrightarrow{p} = U\overleftrightarrow{p'}U^{\dagger}$. In matrix form, it is expressed as:

$$\begin{pmatrix} p_{xx} & p_{xy} & p_{xz} \\ p_{xy} & p_{yy} & p_{yz} \\ p_{xz} & p_{yz} & p_{zz} \end{pmatrix} = \begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} p_{x'x'} & p_{x'y'} & p_{x'z'} \\ p_{x'y'} & p_{y'y'} & p_{y'z'} \\ p_{x'z'} & p_{y'z'} & p_{z'z'} \end{pmatrix} \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{29}$$

Therefore:

$$\begin{aligned} I(\theta,\phi) = I_0(\theta)[1 &+ \frac{3}{2}\left(p_{x'}\sin\phi + p_{y'}\cos\phi\right)A_y(\theta) \\ &+ \frac{2}{3}\left(p_{x'z'}\cos\phi - p_{y'z'}\sin\phi\right)A_{xz}(\theta) \\ &+ \frac{1}{6}\left\{\left(p_{x'x'} - p_{y'y'}\right)\cos 2\phi - 2p_{x'y'}\sin 2\phi\right\}\left\{A_{xx}(\theta) - A_{yy}(\theta)\right\} \\ &+ \frac{1}{2}p_{z'z'}A_{zz}(\theta)\Big]. \end{aligned} \tag{30}$$

Therefore, we can monitor the polarization state of the deuterium beam by placing detectors at the same polar angle θ but different azimuthal angles ϕ (0, 90, 180, 270 degrees). This

allows us to measure the components p_x, p_y, p_{xz}, p_{yz}, p_{zz}, p_{xy}, and $p_{xx}-p_{yy}$ of the polarization state:

$$L = I_0\left\{1 + \frac{3}{2}p_{y'}A_y + \frac{2}{3}p_{x'z'}A_{xz} + \frac{1}{6}\left(p_{x'x'} - p_{y'y'}\right)\left(A_{xx} - A_{yy}\right) + \frac{1}{2}p_{z'z'}A_{zz}\right\} \tag{31}$$

$$R = I_0\left\{1 - \frac{3}{2}p_{y'}A_y - \frac{2}{3}p_{x'z'}A_{xz} + \frac{1}{6}\left(p_{x'x'} - p_{y'y'}\right)\left(A_{xx} - A_{yy}\right) + \frac{1}{2}p_{z'z'}A_{zz}\right\} \tag{32}$$

$$U = I_0\left\{1 - \frac{3}{2}p_{x'}A_y + \frac{2}{3}p_{y'z'}A_{xz} - \frac{1}{6}\left(p_{x'x'} - p_{y'y'}\right)\left(A_{xx} - A_{yy}\right) + \frac{1}{2}p_{z'z'}A_{zz}\right\} \tag{33}$$

$$D = I_0\left\{1 + \frac{3}{2}p_{x'}A_y - \frac{2}{3}p_{y'z'}A_{xz} - \frac{1}{6}\left(p_{x'x'} - p_{y'y'}\right)\left(A_{xx} - A_{yy}\right) + \frac{1}{2}p_{z'z'}A_{zz}\right\} \tag{34}$$

$$\frac{L+R+D+U}{4} = I_0(1 + \frac{1}{2}p_{z'z'}A_{zz}) \tag{35}$$

E. The data of the Analysing Power and the Simulations

In the paper [1], Sekiguchi et al. have measured the analyzing power data required for the D(p,EL)D process.

The analyzing power used in the table of paper [1] uses spherical operators, while the previous analysis was done uses the rectangular operators. The transformation between the two sets of basis is given by equations 36 and 37. Therefore, the transformation relationship for the analyzing power between the two coordinate systems is given by equation 38.

$$\begin{aligned}
\tau_{00} &= I \quad (3\times 3 \text{ unit matrix })\\
\tau_{1\pm1} &= \mp\frac{\sqrt{3}}{2}\left(\mathscr{P}_x \pm \mathrm{i}\mathscr{P}_y\right)\\
\tau_{10} &= \sqrt{\frac{3}{2}}\mathscr{P}_z\\
\tau_{2\pm2} &= \frac{1}{2\sqrt{3}}\left(\mathscr{P}_{xx} - \mathscr{P}_{yy} \pm 2\mathrm{i}\mathscr{P}_{xy}\right)\\
\tau_{2\pm1} &= \mp\frac{1}{\sqrt{3}}\left(\mathscr{P}_{xz} \pm \mathrm{i}\mathscr{P}_{yz}\right)\\
\tau_{20} &= \frac{1}{\sqrt{2}}\mathscr{P}_{zz}
\end{aligned} \tag{36}$$

$$
\begin{aligned}
\mathscr{P}_x &= \frac{-1}{\sqrt{3}}(\tau_{11} - \tau_{1-1}) = -\frac{2}{\sqrt{3}}\,\mathrm{Im}\,\tau_{11},\\
\mathscr{P}_y &= \frac{\mathrm{i}}{\sqrt{3}}(\tau_{11} + \tau_{1-1}) = -\frac{2}{\sqrt{3}}\,\mathrm{Re}\,\tau_{11},\\
\mathscr{P}_z &= \sqrt{\frac{2}{3}}\tau_{10} = \sqrt{\frac{2}{3}}\tau_{10},\\
\mathscr{P}_{xy} &= \frac{-\mathrm{i}\sqrt{3}}{2}(\tau_{22} - \tau_{2-2}) = \sqrt{3}\,\mathrm{Im}\,\tau_{22},\\
\mathscr{P}_{xz} &= \frac{-\sqrt{3}}{2}(\tau_{21} - \tau_{2-1}) = -\sqrt{3}\,\mathrm{Re}\,\tau_{21},\\
\mathscr{P}_{yz} &= \frac{\mathrm{i}\sqrt{3}}{2}(\tau_{21} + \tau_{2-1}) = -\sqrt{3}\,\mathrm{Im}\,\tau_{21},\\
\mathscr{P}_{xx} &= \frac{\sqrt{3}}{2}(\tau_{22} + \tau_{2-2}) - \frac{1}{\sqrt{2}}\tau_{20} = \sqrt{3}\,\mathrm{Re}\,\tau_{22} - \frac{1}{\sqrt{2}}\tau_{20},\\
\mathscr{P}_{yy} &= -\frac{\sqrt{3}}{2}(\tau_{22} + \tau_{2-2}) - \frac{1}{\sqrt{2}}\tau_{20} = -\sqrt{3}\,\mathrm{Re}\,\tau_{22} - \frac{1}{\sqrt{2}}\tau_{20},\\
\mathscr{P}_{zz} &= \sqrt{2}\tau_{20} = \sqrt{2}\tau_{20}.
\end{aligned}
\tag{37}
$$

$$
\begin{aligned}
A_x &= \frac{-1}{\sqrt{3}}(T_{11} - T_{1-1}) = -\frac{2}{\sqrt{3}}\,\mathrm{Im}\,T_{11},\\
A_y &= \frac{\mathrm{i}}{\sqrt{3}}(T_{11} + T_{1-1}) = -\frac{2}{\sqrt{3}}\,\mathrm{Re}\,T_{11},\\
A_z &= \sqrt{\frac{2}{3}}T_{10} = \sqrt{\frac{2}{3}}T_{10},\\
A_{xy} &= \frac{-\mathrm{i}\sqrt{3}}{2}(T_{22} - T_{2-2}) = \sqrt{3}\,\mathrm{Im}\,T_{22},\\
A_{xz} &= \frac{-\sqrt{3}}{2}(T_{21} - T_{2-1}) = -\sqrt{3}\,\mathrm{Re}\,T_{21},\\
A_{yz} &= \frac{\mathrm{i}\sqrt{3}}{2}(T_{21} + T_{2-1}) = -\sqrt{3}\,\mathrm{Im}\,T_{21},\\
A_{xx} &= \frac{\sqrt{3}}{2}(T_{22} + T_{2-2}) - \frac{1}{\sqrt{2}}T_{20} = \sqrt{3}\,\mathrm{Re}\,T_{22} - \frac{1}{\sqrt{2}}T_{20},\\
A_{yy} &= -\frac{\sqrt{3}}{2}(T_{22} + T_{2-2}) - \frac{1}{\sqrt{2}}T_{20} = -\sqrt{3}\,\mathrm{Re}\,T_{22} - \frac{1}{\sqrt{2}}T_{20},\\
A_{zz} &= \sqrt{2}T_{20} = \sqrt{2}T_{20}.
\end{aligned}
\tag{38}
$$

The analysing power is shown in figure 2.

The differential cross section is shown in figure:3

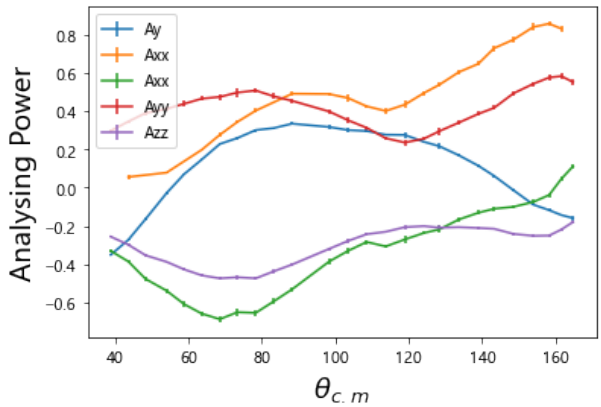


FIG. 2: The relationship between the analyzing power and the center-of-mass scattering angle.

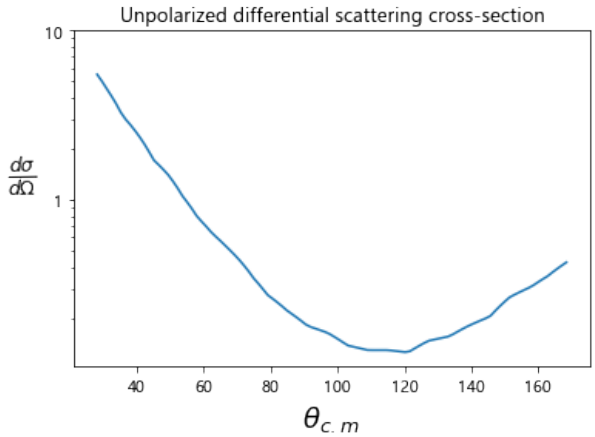


FIG. 3: The unpolarized differential cross section.

We will perform the experiment with a deuteron beam intensity of 1.6×10^{-3} pnA with a 1000 mg/cm^2 thick polyethylene (CH2) target in the scattering chamber. According to the reference [2], the detector angles can be adjusted in the range of 7.5 to 70° in the laboratory frame, with a solid angle coverage of $\Delta\theta_{\text{lab}} = \pm 1°$. This corresponds to a solid angle of $\Delta\Omega = \pi\Delta\theta^2_{\text{lab}} = 9.6 \times 10^{-4}$ sr .

The differential scattering cross section in the laboratory system is shown in Figure 4. Additionally, the variation of the analyzing power A_{zz} with the scattering angle in the laboratory system is depicted in Figure 5. In order to achieve better significance, the analyzing power A_{zz} needs to be relatively large and exhibit minimal variation with angle. It is evident that the optimal angle for the detector is $\theta_{\text{lab}} = 22°$.

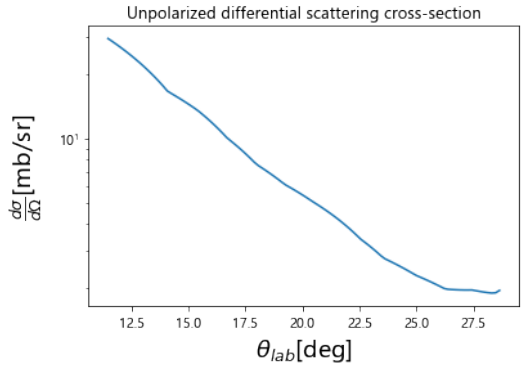


FIG. 4: The elastic scattering cross section of unpolarized 190 MeV/nucleon deuterons on protons in the laboratory system.

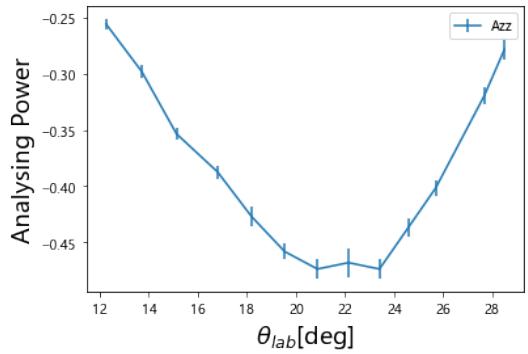


FIG. 5: The distribution of the analyzing power A_{zz} with the laboratory scattering angle .

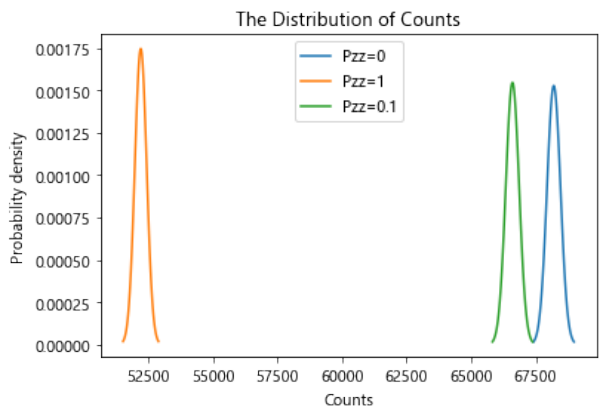


FIG. 6: The probability distribution of the total detector counts under different polarization rates .

The probability of a deuterium nucleus scattering onto four detectors is given by $p =$

$4\frac{d}{M_{\mathrm{CH2}}}N_A\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\Big|\mathrm{cm}\Delta\Omega$. The total counts in the detectors follow a binomial distribution, which can be approximated using the central limit theorem. The probability distribution of counts obtained from monitoring for three hours is shown in Figure II E. Comparing the mean count with $P_{zz} = 0.1$ and the count distribution in the unpolarized state, there is a difference of 1598, while $3\sigma_1 + 3\sigma_2$ is 1558. Therefore, the tensor polarization rate can be monitored with an accuracy of 10 which matches the fluctuations of the polarization rate of the polarized deuteron beam generated by the apparatus over the experimental period of 9 days.

In addition to placing four detectors at different azimuthal angles for a given polar angle, we can also use the count ratio at two polar angles to monitor the polarization rate. This approach helps to mitigate the influence of beam intensity variations.

III. CONCLUSION

The polarization of deuterons with isospin vector polarization effect provides a clean probe of the symmetry energy at around 0.5 times saturation density. Deuterons polarized along the beam direction exhibit distinct angular differences in the protons and neutrons produced from their breakup in target-nucleus scattering. In this work, we address the detection of tensor polarization in the direction of the deuteron beam. We plan to submit a beam proposal to RIKEN in Japan this year. The experiment will have a duration of nine days, allowing us to determine the polarization state of the deuteron beam every three hours. The results of the experiment will provide insights into the stiffness of the symmetry energy in the region of 0.5 times saturation density.

[1] K. Sekiguchi, H. Witała, T. Akieda, D. Eto, H. Kon, Y. Wada, A. Watanabe, S. Chebotaryov, M. Dozono, J. Golak, et al., Physical Review C 96, 064001 (2017).

[2] K. Sekiguchi, Y. Wada, J. Miyazaki, H. Witała, M. Dozono, U. Gebauer, J. Golak, H. Kamada, S. Kawase, Y. Kubota, et al., Physical Review C 89, 064007 (2014).